

ReclaimIQ

Benefit-Underutilization Analytics

Putting a **rupee figure** on unclaimed card benefits — and turning it into **measurable** engagement.

CodeStreet 2026 · American Express India

The problem

- Amex India cards bundle ₹1,000s of value: credits, lounge, protection, **Offers**, **milestones**, **reward points**.
- **Issuers** can't measure how much value goes **unclaimed** → benefit ROI is a guess.
- **Members** don't see what they're missing → perceived value drops → **churn**.

The value is already paid for. The gap is **visibility + activation**, not entitlement.

Grounded in real Amex India products

Card	Fee	What we model
Platinum Charge	₹66,000	Unlimited lounge · ₹40K credits · protection · ₹20L voucher
Platinum Travel	₹5,000	Milestones ₹1.9L/₹4L/₹7L → MR + Taj voucher (2026 structure)
MRCC	₹4,500	Monthly milestones (4×₹1,500 · ₹20K spend, enrolment)
SmartEarn	₹495	10X/5X online accelerators · ₹500 voucher milestones

Real products; synthetic, privacy-safe member data.

The opportunity (synthetic book: 400 members, ~200K transactions)

₹50.1 L

annualized **unclaimed** value · ₹12,519 / member · portfolio utilization ≈63%

- **114** members below **25%** utilization — the priority segment.
- Leakage concentrates in **lounge (₹33.5L)**, then **milestones (₹10.6L)**, **protection (₹2.2L)**, **credits (₹2.2L)**, **offers (₹1.6L)**.
- **₹42.0 L** of Membership Rewards value sits idle in accounts.

How we quantify the gap — six families, transparent by design

Family	Unclaimed logic
Credits	<code>min(cap, qualifying spend NET OF REFUNDS)</code> per cycle
Lounge	reachable travel days – visits (<i>detected from in-lounge spend + passes</i>)
Protection	<code>coverage × protected spend</code> – claims (value-at-risk)
Offers	spent at merchant but offer not activated = missed
Milestones	near-miss / lost-to-non-enrolment vs. spend threshold
Points	balance × redemption range (₹0.25 → ₹1.50 / pt)

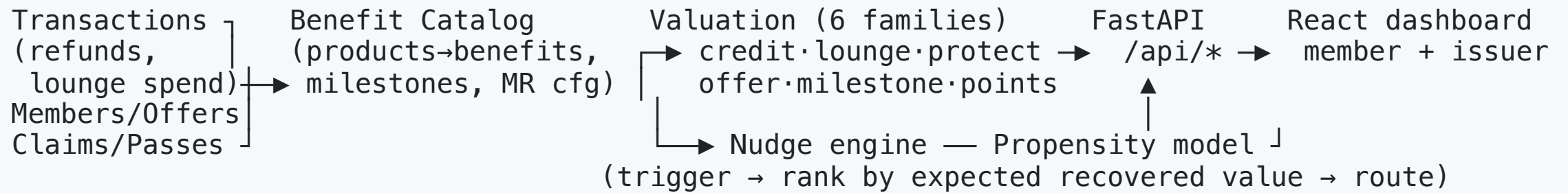
Every rupee traces to an **explicit rule** → defensible to members *and* stakeholders.

Solution: ReclaimIQ — one engine, two audiences

- **Card Member view** — *"You're ₹54,647 away from unlocking a ₹5,000 milestone."*
"You spent ₹5,358 at Zomato — the Amex Offer was never activated."
- **Issuer Portfolio view** — book-level unclaimed value, utilization, points at stake, and nudge ROI.

Same **defensible math** powers the member nudge *and* the business case.

Architecture



Stack: **Python (FastAPI, pandas, scikit-learn)** + **React**. Warehouse/stream ready.

Nudge engine — right message, right moment

1. **Trigger** on patterns: un-activated **offer**, **near-miss milestone**, unused **lounge** on travel days, idle **points**, **expiring** credit.
2. **Rank** by **expected recovered value** = ₹ gap × P(convert | nudge).
3. **Route** by intent (urgent → push · transactional → in-app · awareness → email) + timing.

Optimizes for **recovered rupees**, not clicks.

The model & measurable uplift (Task 5)

- Gradient-boosted **propensity model** (+ transparent logistic fallback) on a labelled log.
- **AUC ≈ 0.64 , top-decile lift $\approx 1.54\times$** over a **25.6%** base conversion rate.

4.29x

more value recovered vs. a random/blanket campaign of the same size.

(Synthetic label is drawn from a known formula the model re-learns — demonstrates the pipeline, not a real effect size.)

Why it wins

- **Rupee-denominated & explainable** gap across **six** benefit families.
- **Activation-aware** Offers & **near-miss** milestones — models the *state* that causes leakage.
- **Points as a value range** — surfaces the 6× statement-credit-vs-transfer gap.
- **Expected-value targeting** — spends nudge budget where it recovers most.
- **Dual-audience, single engine** — member value = issuer ROI, always consistent.

Scope & future work

- **Built:** refund netting, lounge-from-transactions, Amex Offers activation, milestones, points valuation — all grounded in real Amex India products.
- **Future (documented, not built):** partner-vs-non-partner **price comparison** (needs historical real-time pricing); **cross-card** optimization (out of PS scope — PS is per-member across credits/lounge/protection).

Feasibility · Scalability · Impact

- **Feasible:** standard stack; engine is a pure fn of (transactions, entitlements).
- **Scalable:** $O(\text{transactions})$, per-member parallel → Spark / Snowflake / BigQuery; catalog is *config, not code*.
- **Impact:** retention lift, benefit-ROI clarity, efficient engagement spend.

Thank you

ReclaimIQ — the value is already there. We help members **reclaim** it, and issuers **prove** it.

Working prototype + code in the repo. All member data synthetic & privacy-safe.